

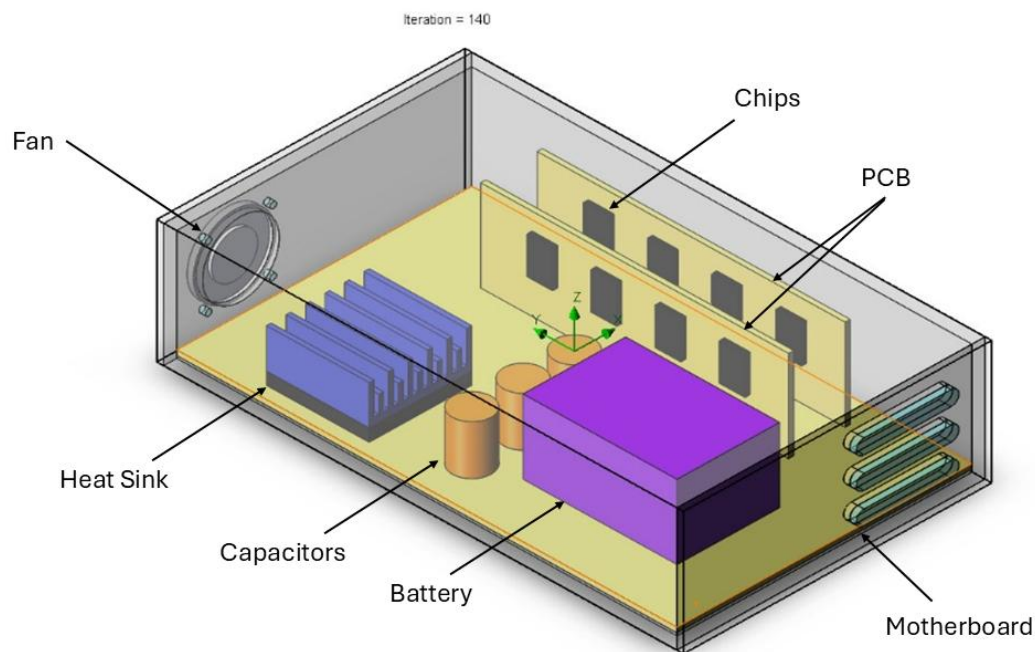
CFD Simulation of an electronic circuit board



Pratik Malu

SOLIDWORKS Flow Simulation Project Report

October 29, 2025



Isometric

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1 General Information

Objective of the simulation: CFD Simulation of an electronic circuit board, with capacitors, chips, fan, battery, PCB boards. Reference from a YouTube Video

1.1 Analysis Environment

Software Product: Flow Simulation 2025 SP0.0. Build: 6475
CPU Type: AMD Ryzen 7 5800H with Radeon Graphics
CPU Speed: 3201 MHz
RAM: 15753 MB / 7211 MB
Operating System: Windows 11 (or higher) (Version 10.0.26200)

1.2 Model Information

Model Name: assembly.SLDASM
Project Name: Electronics CFD

1.3 Project Comments:

Unit System: USA
Analysis Type: Internal

1.4 Size of Computational Domain

<i>Size</i>	
X min	-3.006 in
X max	3.006 in
Y min	-5.130 in
Y max	5.135 in
Z min	-1.253 in
Z max	1.253 in
X size	6.012 in
Y size	10.265 in
Z size	2.505 in

1.5 Simulation Parameters

1.5.1 Mesh Settings

1.5.1.1 Basic Mesh

<i>Basic Mesh Dimensions</i>	
Number of cells in X	16
Number of cells in Y	26
Number of cells in Z	6

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1.5.1.2 Analysis Mesh

Total Cell count: 82052
Fluid Cells: 42084
Solid Cells: 39968
Partial Cells: 18531
Trimmed Cells: 0

1.5.1.3 Additional Physical Calculation Options

Heat Transfer Analysis: Fluid Flow: OnConduction: On
Flow Type: Laminar and turbulent
Time-Dependent Analysis: Off
Gravity: Off
Radiation: Off
Humidity: Off
Default Wall Roughness: 0 microinch

1.5.2 Material Settings

Material Settings

Fluids

[Air](#)

Solids

[Steel Stainless 321](#)

[MAT PCB](#)

[Package](#)

[Glass](#)

[Aluminum](#)

1.5.3 Initial Conditions

Initial Conditions

Thermodynamic parameters	Static Pressure: 14.695947 lbf/in ² Temperature: 50.00 °F
Velocity parameters	Velocity vector Velocity in X direction: 0 ft/s Velocity in Y direction: 0 ft/s Velocity in Z direction: 0 ft/s
Solid parameters	Default material: Steel Stainless 321 Initial solid temperature: 68.09 °F
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.03 in

1.5.4 Boundary Conditions

Fans

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External Inlet Fan 1

Type	External Inlet Fan
Fan curve	Papst 412
	Face<1>@in lid<1>
Coordinate system	Face Coordinate System
Reference axis	X
Flow parameters	Inlet flow vector direction: Normal to face
Thermodynamic parameters	Environment pressure: 14.695947 lbf/in ² Temperature: 50.00 °F
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.03 in
Toggle	On

Boundary Conditions

Environment Pressure 1

Type	Environment Pressure
Faces	Face<1>@out lid<2> Face<1>@out lid<3> Face<1>@out lid<1>
Coordinate system	Global Coordinate System
Reference axis	X
Thermodynamic parameters	Environment pressure: 14.695947 lbf/in ² Temperature type: Temperature of initial components Temperature: 50.00 °F
Turbulence parameters	Turbulence intensity and length Intensity: 2.00 % Length: 0.03 in
Boundary layer parameters	Boundary layer type: Turbulent

1.5.5 Volumetric Heat Sources

Heat Volume Sources

Capacitors Temperature

Components	capacitor<3> capacitor<1> capacitor<2>
Coordinate system	Global Coordinate System
Reference axis	X
Source type	Temperature
Temperature	100.00 °F

Small Chip Heat Generation Rate

Components	smallchip<6> smallchip<8> smallchip<7> smallchip<1> smallchip<2> smallchip<3> smallchip<4> smallchip<5>
Coordinate system	Global Coordinate System
Reference axis	X
Source type	Heat Generation Rate

Fluid Flow Simulation Report

Heat generation rate	4.000 W
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Battery Temperature

Components	power<1>
Coordinate system	Global Coordinate System
Reference axis	X
Source type	Temperature
Temperature	120.00 °F

1.5.6 Engineering Goals

Goals

Global Goals

GG Average Static Pressure 1

Type	Global Goal
Goal type	Static Pressure
Calculate	Average value
Coordinate system	Global Coordinate System
Use in convergence	On

GG Average Temperature (Fluid) 2

Type	Global Goal
Goal type	Temperature (Fluid)
Calculate	Average value
Coordinate system	Global Coordinate System
Use in convergence	On

Surface Goals

Small Chip Max T-Solid

Type	Surface Goal
Goal type	Temperature (Solid)
Calculate	Maximum value
Faces	smallchip<6> smallchip<8> smallchip<7> smallchip<1> smallchip<2> smallchip<3> smallchip<4> smallchip<5>
Coordinate system	Global Coordinate System
Use in convergence	On

SG Average Static Pressure 2

Type	Surface Goal
Goal type	Static Pressure
Calculate	Average value
Faces	Face<1>@in lid<1>
Coordinate system	Global Coordinate System
Use in convergence	On

SG Mass Flow Rate 3

Fluid Flow Simulation Report

Type	Surface Goal
Goal type	Mass Flow Rate
Faces	Face<1>@out lid<2> Face<1>@out lid<3> Face<1>@out lid<1>
Coordinate system	Global Coordinate System
Use in convergence	On

Volume Goals

Main Chip Max T-Solid

Type	Volume Goal
Goal type	Temperature (Solid)
Calculate	Maximum value
Components	mainchip<1>
Coordinate system	Global Coordinate System
Use in convergence	On

1.6 Analysis Time

Calculation Time: 76 s

Number of Iterations: 140

Warnings:

2 Results

2.1 Analysis Goals

Goals

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
GG Average Static Pressure 1	lbf/in ²	14.696985	100	9.29588866e-06	1.79419397e-06	On
GG Average Temperature (Fluid) 2	°F	62.84	100	0.355008418	0.128546505	On
Small Chip Max T-Solid	°F	94.18	100	0.449882514	0.426138807	On
SG Average Static Pressure 2	lbf/in ²	14.696345	100	5.62276632e-05	4.22148032e-06	On
SG Mass Flow Rate 3	lb/s	-0.0074	100	0.000131788998	3.5287213e-06	On
Main Chip Max T-Solid	°F	94.78	100	0.323016198	0.173447018	On

2.2 Global Min-Max-Table

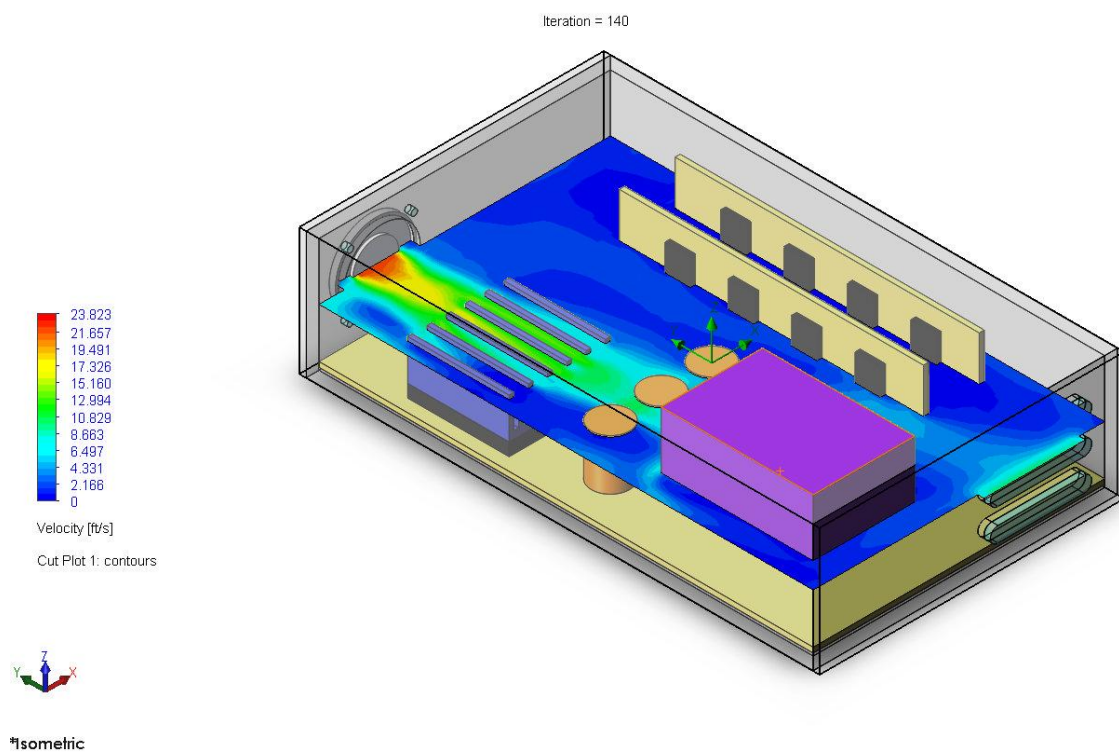
Min/Max Table

Name	Minimum	Maximum
Density (Fluid) [lb/ft ³]	0.069	0.078
Density (Solid) [lb/ft ³]	69.919	505.665
Pressure [lbf/in ²]	14.695511	14.698961
Temperature [°F]	50.00	120.00
Temperature (Fluid) [°F]	50.00	120.00
Temperature (Solid) [°F]	54.92	120.00
Velocity [ft/s]	0	23.823
Velocity (X) [ft/s]	-11.022	10.195
Velocity (Y) [ft/s]	-21.488	9.154
Velocity (Z) [ft/s]	-9.999	10.464
Mach Number []	0	0.02
Velocity RRF [ft/s]	0	23.823
Velocity RRF (X) [ft/s]	-11.022	10.195
Velocity RRF (Y) [ft/s]	-21.488	9.154
Velocity RRF (Z) [ft/s]	-9.999	10.464
Vorticity [1/s]	0.35	1520.06
Relative Pressure [lbf/in ²]	-0.000436	0.003014
Shear Stress [lbf/in ²]	0	0.000052
Bottleneck Number []	0	1.0000000

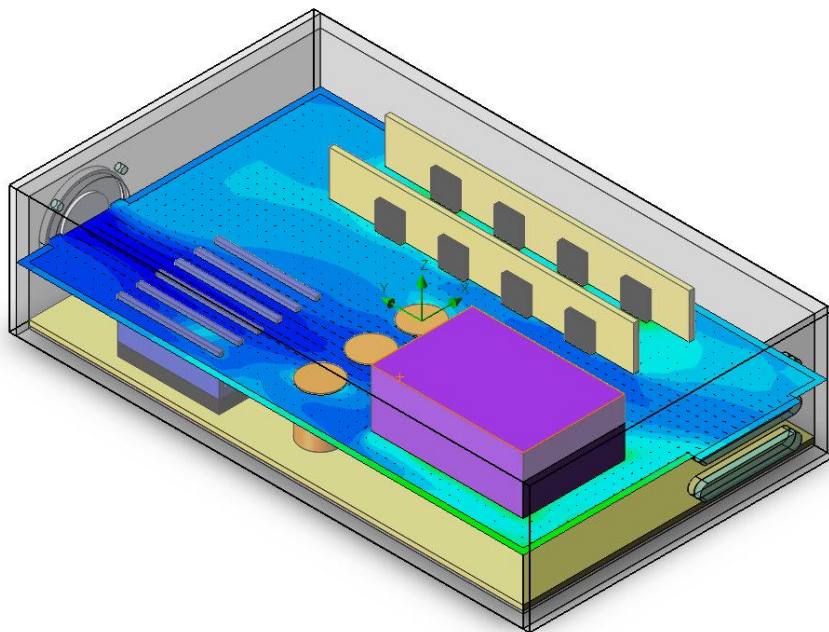
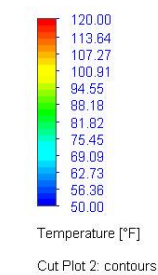
Fluid Flow Simulation Report

Heat Transfer Coefficient [W/m ² /K]	1.5406e-05	171.0548
Overheat above Melting Temperature [°F]	-2511.139	-142.330
ShortCut Number []	0	1.0000000
Surface Heat Flux [W/m ²]	-400097.1756	400097.1756
Surface Heat Flux (Conductive) [W/m ²]	-400097.1756	400097.1756
Surface Heat Flux (Convective) [W/m ²]	-296.6418	3597.5857
Total Enthalpy Flux [W/m ²]	-1180153.7901	2370446.5416
Volumetric Heat Generation Rate [W/m ³]	0	813649.9213
Acoustic Power [W/m ³]	0	2.6258e-11
Acoustic Power Level [dB]	0	14.19

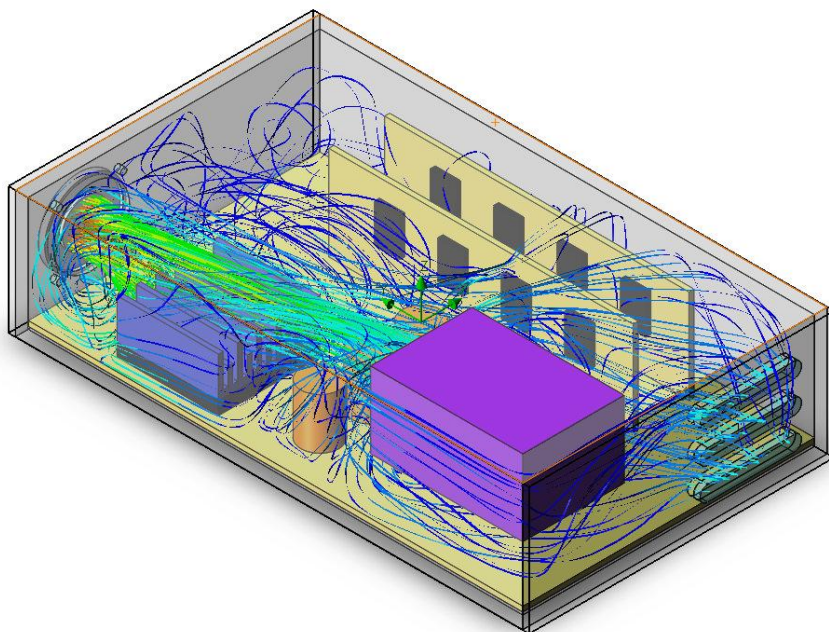
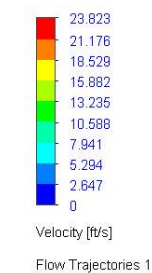
2.3 Results



Velocity Cut Plot on showcasing velocity contours



Temperature Cut Plot on showcasing temperature contours



Velocity Flow trajectory in the enclosed domain

3 Appendix

3.1 Material Data

Engineering Database

Gases

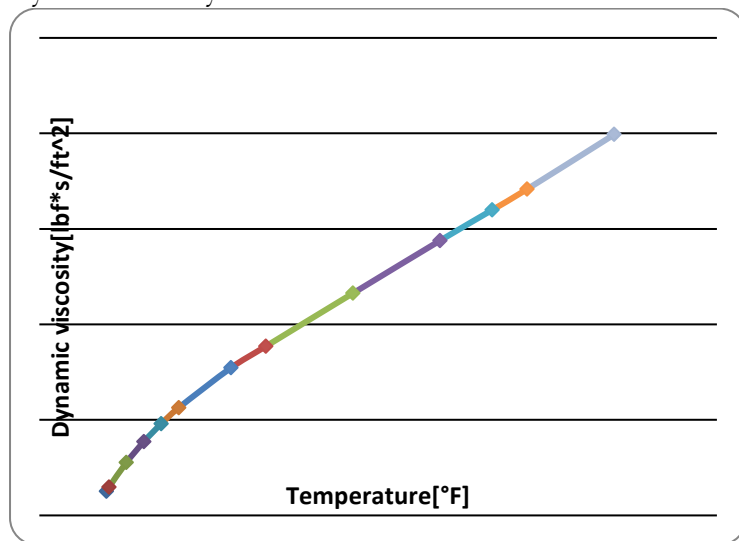
Air

Path: Gases Pre-Defined

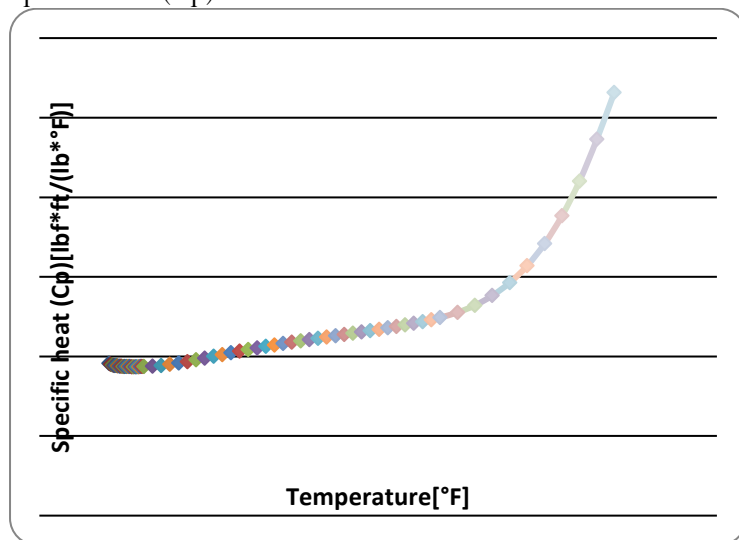
Specific heat ratio (C_p/C_v): 1.399

Molecular mass: 0.0638 lb/mol

Dynamic viscosity

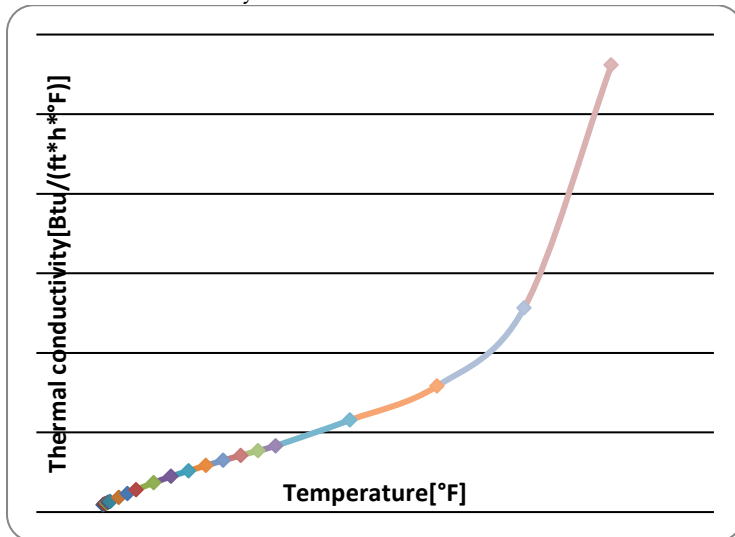


Specific heat (C_p)



Fluid Flow Simulation Report

Thermal conductivity



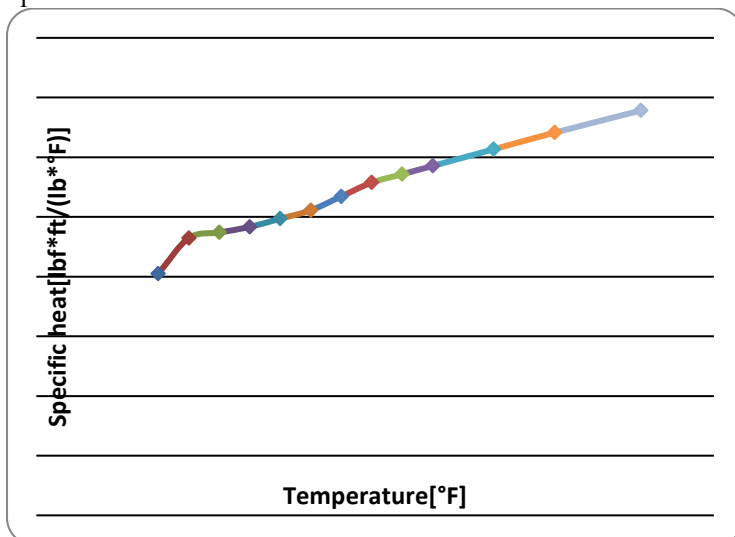
Solids

Steel Stainless 321

Path: Solids Pre-Defined\Alloys

Density: 505.665 lb/ft³

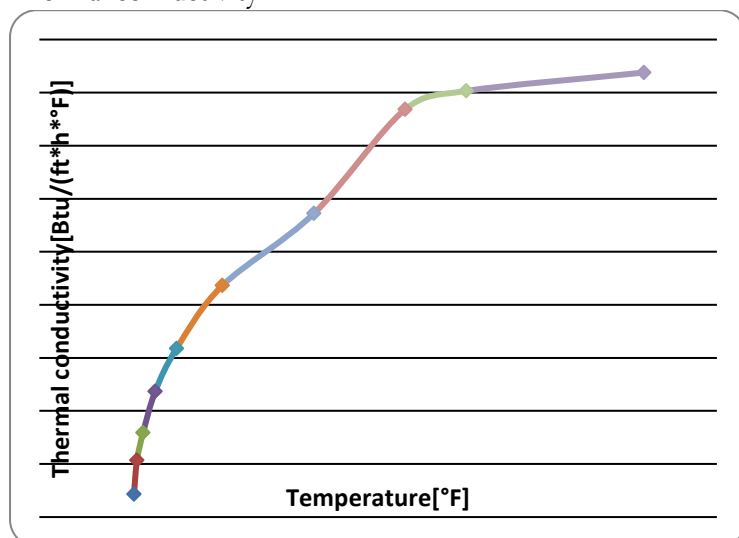
Specific heat



Conductivity type: Isotropic

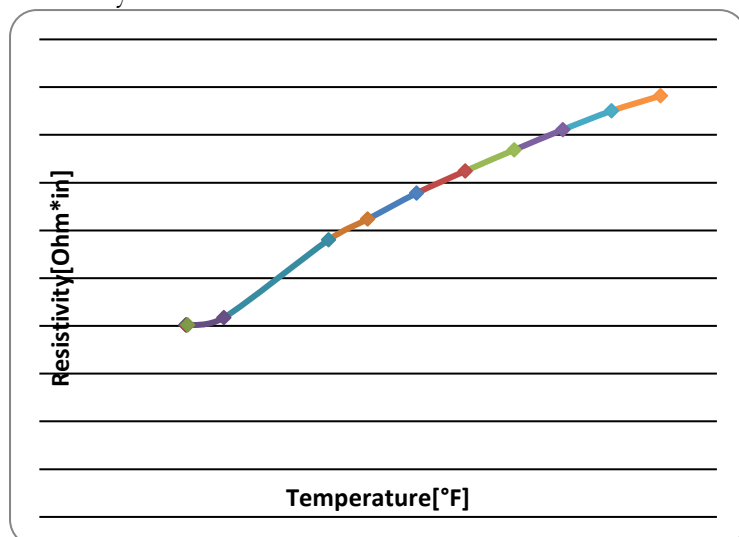
Fluid Flow Simulation Report

Thermal conductivity



Electrical conductivity: Conductor

Resistivity



Radiation properties: No

Melting temperature: Yes

Temperature: 2570.00 °F

MAT PCB

Path: Solids User Defined

Density: 69.919 lb/ft³

Specific heat: 260.2 lbf*ft/(lb*°F)

Conductivity type: Isotropic

Thermal conductivity: 5.7779 Btu/(ft*h*°F)

Electrical conductivity: Dielectric

Radiation properties: No

Melting temperature: Yes

Temperature: 242.33 °F

Package

Path: Solids User Defined\New Folder

Density: 124.856 lb/ft³

Specific heat: 22.3 lbf*ft/(lb*°F)

Fluid Flow Simulation Report

Conductivity type: Isotropic

Thermal conductivity: 0.2311 Btu/(ft*h*°F)

Electrical conductivity: Dielectric

Radiation properties: No

Melting temperature: Yes

Temperature: 242.33 °F

Glass

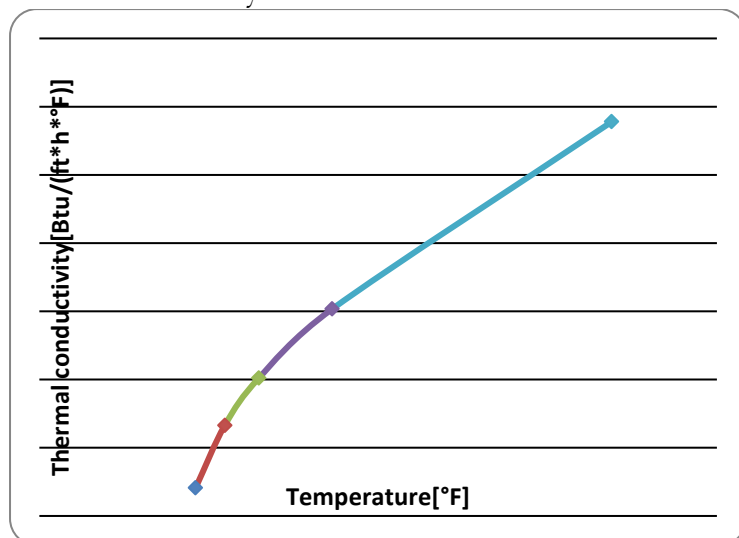
Path: Solids Pre-Defined\Glasses and Minerals

Density: 162.312 lb/ft³

Specific heat: 124.5 lbf*ft/(lb*°F)

Conductivity type: Isotropic

Thermal conductivity



Electrical conductivity: Dielectric

Radiation properties: No

Melting temperature: Yes

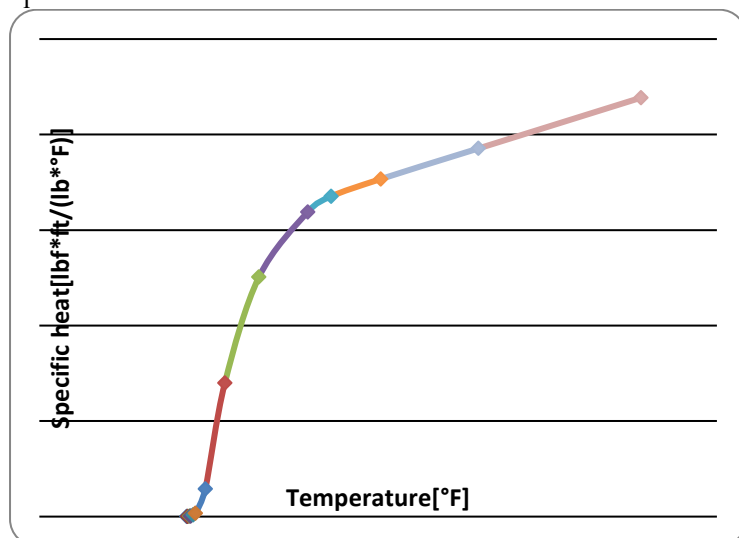
Temperature: 1472.00 °F

Aluminum

Path: Solids Pre-Defined\Metals

Density: 167.862 lb/ft³

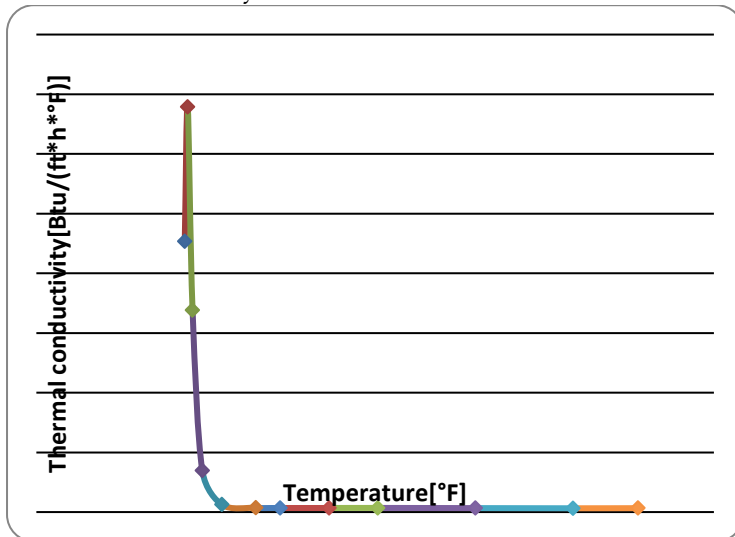
Specific heat



Conductivity type: Isotropic

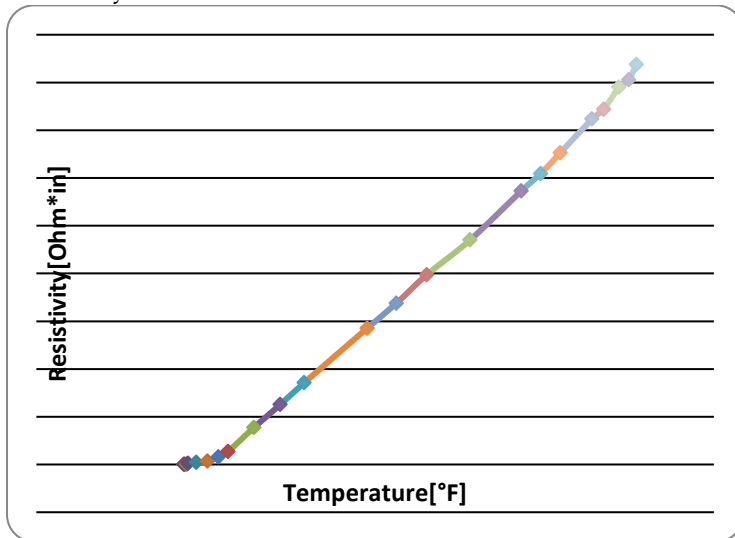
Fluid Flow Simulation Report

Thermal conductivity



Electrical conductivity: Conductor

Resistivity



Radiation properties: No

Melting temperature: Yes

Temperature: 1220.45 °F